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In [1]: import pandas as pd
import numpy as np

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

import re
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In [2]: df = pd.read_csv("netflix_titles.csv")

df.head()
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Out[2]:
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	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021

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In [3]: df.info()
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<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        6173 non-null   object
4   cast            7982 non-null   object
5   country         7976 non-null   object
6   date_added      8797 non-null   object
7   release_year    8807 non-null   int64
8   rating          8803 non-null   object
9   duration        8804 non-null   object
10  listed_in       8807 non-null   object
11  description     8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
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In [4]: df = df[['title', 'director', 'cast', 'listed_in', 'description']]
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In [5]: df.fillna('', inplace=True)
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In [6]: def clean_text(text):
        text = text.lower()
        text = re.sub(r'^a-zA-Z\s', '', text)
        return text
```

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In [7]: df['soup'] = (
        df['director'] + ' ' +
        df['cast'] + ' ' +
        df['listed_in'] + ' ' +
        df['description']
    )

df['soup'] = df['soup'].apply(clean_text)
```

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In [8]: tfidf = TfidfVectorizer(stop_words='english')

tfidf_matrix = tfidf.fit_transform(df['soup'])
```

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In [9]: cosine_sim = cosine_similarity(tfidf_matrix, tfidf_matrix)
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In [10]: indices = pd.Series(df.index, index=df['title']).drop_duplicates()
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In [11]: def recommend_movies(title, num_recommendations=5):
        if title not in indices:
            return "Movie not found in database."

        idx = indices[title]

        sim_scores = list(enumerate(cosine_sim[idx]))
        sim_scores = sorted(sim_scores, key=lambda x: x[1], reverse=True)

        sim_scores = sim_scores[1:num_recommendations+1]

        movie_indices = [i[0] for i in sim_scores]
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return df['title'].iloc[movie_indices]
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In [12]: recommend_movies("Inception")
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Out[12]: 3452      Peaky Blinders  
        6272      Before the Flood  
        6376              Brick  
        808       Sniper: Legacy  
        161       Mars Attacks!  
        Name: title, dtype: object
```

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In [13]: recommend_movies("The Matrix")
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Out[13]: 8415      The Matrix Reloaded  
        8416      The Matrix Revolutions  
        6501              Cloud Atlas  
        7151      Jupiter Ascending  
        636       The Ice Road  
        Name: title, dtype: object
```

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In [ ]:
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